

Computer Network Practice: *ARP*

CC Lab.

Taewook Kim (TA)

Computer Communication Lab.

Dept. of Computer Engineering,

Chungbuk National University





Lecture Plan

Week	Date	Contents
1	9.1	-
2	9.8	Introduction & Ping
3	9.15	Inter-Process Communication
4	9.22	File Transfer 1
5	9.29	File Transfer 2
6	10.6	ARP
7	10.13	GARP, RARP
8	10.20	(middle test)
9	10.27	Internet Protocol Routing I & Static Router
10	11.3	Internet Protocol Routing II & Static Router
11	11.10	Internet Protocol Routing III & Static Router
12	11.17	Network Address Translation I
13	11.24	Network Address Translation II
14	12.1	Network Address Translation III
15	12.8	(final test)





Contents

- ◆ Address Resolution Protocol
 - ❖ Demultiplexing
- ◆ ARP Message Format
 - ❖ Description of ARP Field
- ◆ ARP Cache Table
- ◆ ARP Procedure
 - ❖ Example





Contents

- ◆ ARP Architecture
- ◆ Network Topology for Test
- ◆ Basic Operation of ARP
 - ❖ Request Operation
 - ❖ Reply Operation
 - ❖ ARP Basic Operation
- ◆ Notice





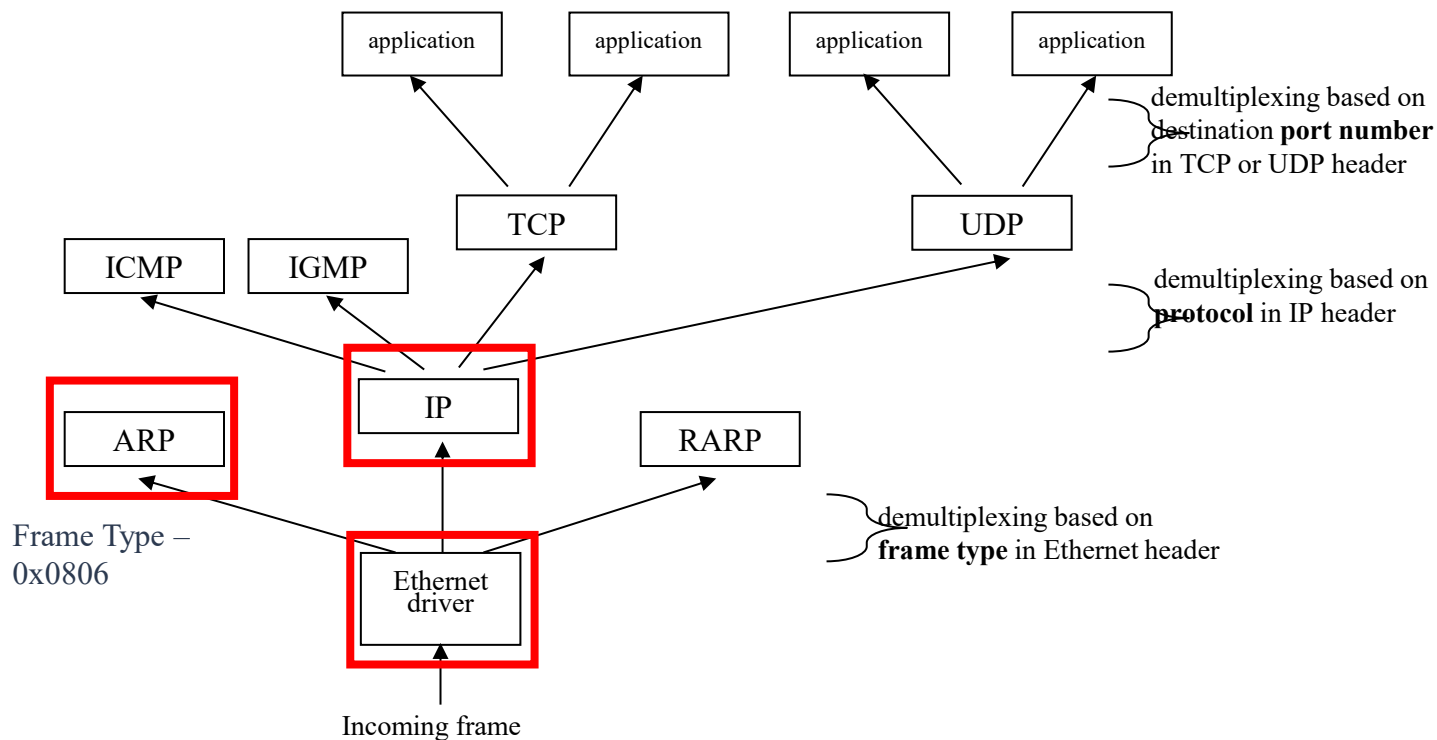
Address Resolution Protocol

- ◆ What is the reason it needs ARP in LAN?
 - ❖ When Ethernet frames are sent from one host on a LAN to another
 - ◆ it is the 48-bit Ethernet address that determines for which interface the frame is destined.
 - ❖ Mapping between the two different types of addresses
 - ◆ 32-bit IP address and whatever type of address the data link uses.
 - ❖ Provides a dynamic mapping from an IP address to the corresponding hardware address.





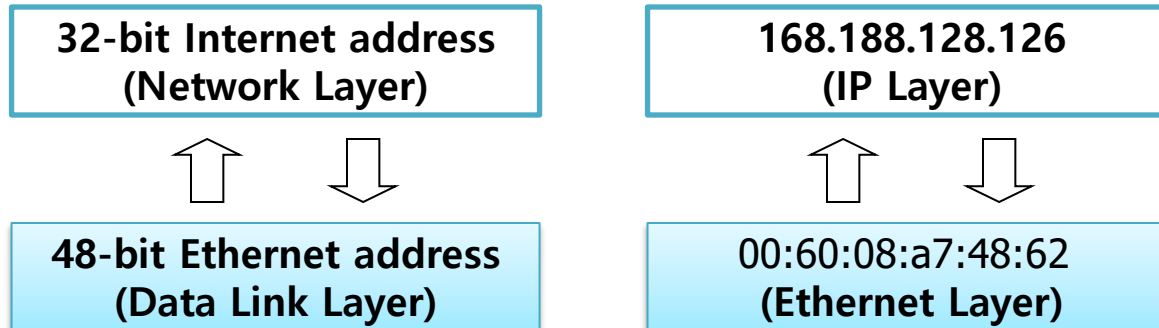
Demultiplexing



The demultiplexing of a received Ethernet frame



Address Resolution Protocol



Device Address

IP Address

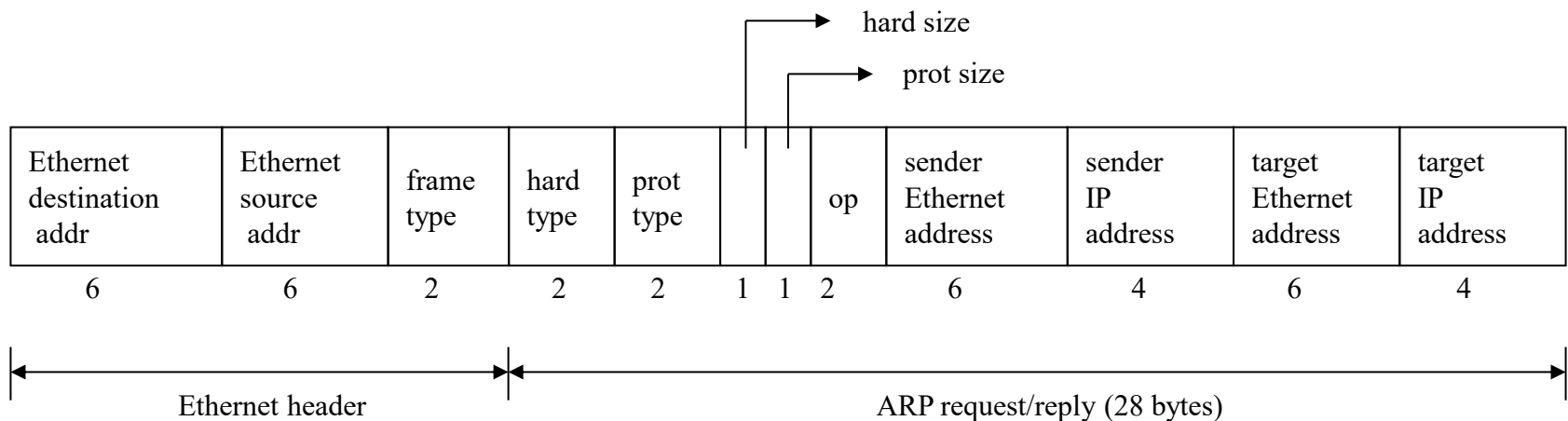
Ethernet

hme0	168.188.129.72	
	00:60:08:a7:43:9a	
hme0	rtislab.cs.chungbuk.ac.kr	08:00:20:7a:91:f6
hme0	java.cs.chungbuk.ac.kr	00:60:08:a7:48:7d
hme0	multimedia.cs.chungbuk.ac.kr	00:60:08:a7:47:78
hme0	168.188.128.126	
	00:60:08:a7:48:62	
hme0	168.188.129.126	
	00:01:02:57:8b:ae	
hme0	168.188.129.123	
	00:60:08:a7:47:7a	
hme0	parfait.cs.chungbuk.ac.kr	00:60:08:a7:4b:72
hme0	168.188.128.96	
	00:c0:26:5c:01:42	
hme0	168.188.128.97	
	00:c0:26:5b:89:13	
hme0	cyberman.cs.chungbuk.ac.kr	00:60:08:a7:48:1e

Find
Path
to
Destination



ARP Message Format



Format of ARP request or reply packet when used on an IP over Ethernet





Description of ARP Field

◆ Ethernet Destination address Field

❖ Broadcast address

◆ All Ethernet interfaces on the cable receive these frames.

◆ Ethernet Source address Field

❖ Ethernet Address of Sender

◆ Frame type Field

❖ For an ARP request or an ARP reply, this field is **0x0806**

◆ Hard type Field

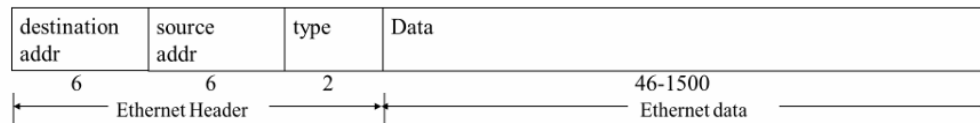
❖ Type of hardware address.

❖ In case of Ethernet - this field is **1**

◆ Prot type Field

❖ Type of protocol address being mapped

❖ In case of IP - this field is **0x0800**



Hardware type (2 bytes)		Protocol type (2 bytes)
Hardware address length (1 byte)	Protocol address length (1 byte)	Operation code (2 bytes)
Source hardware address*		
Source protocol address*		
Target hardware address*		
Target protocol address*		



Description of ARP Field

◆ Hard size & Prot size Field

- ❖ Specify the sizes in bytes of the hardware addresses and the protocol addresses
- ❖ In case of Ethernet - 6 bytes
- ❖ In case of IP - 4 bytes

◆ Op(operation) Field

- ❖ ARP request (a value of 1)
- ❖ ARP reply (a value of 2)
- ❖ RARP request (a value of 3)
- ❖ RARP reply (a value of 4)

◆ Sender's hardware address & Sender's Protocol address

◆ Target hardware address & target Protocol address

Hardware type (2 bytes)		Protocol type (2 bytes)
Hardware address length (1 byte)	Protocol address length (1 byte)	Operation code (2 bytes)
Source hardware address*		
Source protocol address*		
Target hardware address*		
Target protocol address*		





ARP Cache Table

◆ Maintains the latest mappings from Internet address to hardware address.

◆ Timeout

❖ Completed Entry

◆ The normal expiration time of an entry in the cache is **20 minutes** from the time the entry was created.

❖ Incomplete Entry

◆ The network ID and subnet ID are that of local Ethernet, but there is no host with the specified host ID.

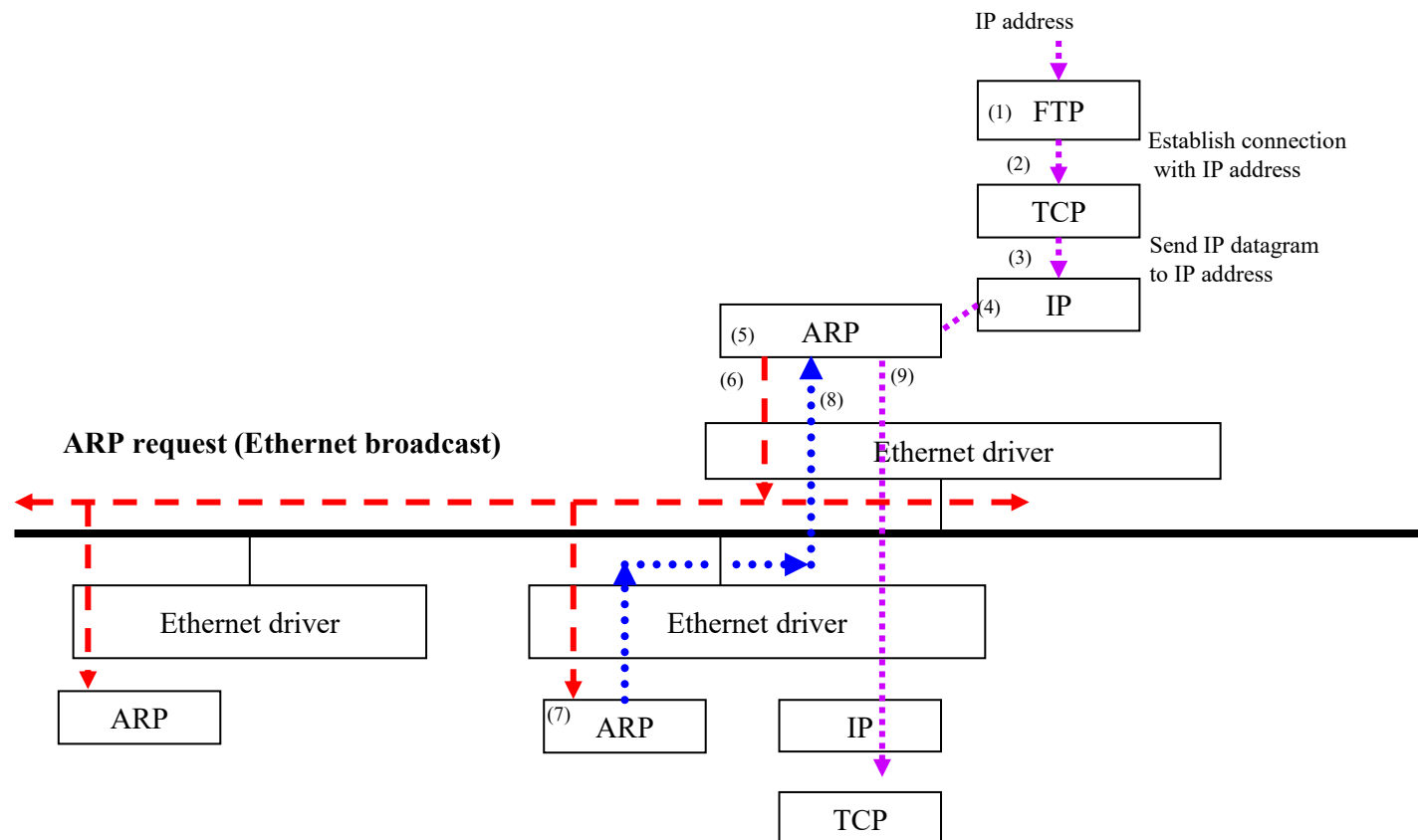
◆ Down or non-existent?

◆ The incomplete expiration time of an entry in the cache is **3 minutes** from the time the entry was created.





ARP Procedure





Example

In case of execute "telnet 168.188.129.2" in cclab03 computer.

Question?

- What is the Ethernet Address of Host that have IP Address 168.188.129.2 ?

Cclab03.cs.cbnu.ac.kr

cs.cbnu.ac.kr



168.188.129.63/24
00:60:08:A7:47:63

Ethernet



168.188.129.2/24
08:00:20:81:28:BD





Example

1. Searching ARP Cache in 168.188.129.63

Interface	IP Address	Ethernet Address	Status
hme0	java.cs.chungbuk.ac.kr	00:60:08:a7:48:7d	Complete
hme0	168.188.129.126	00:60:08:a7:48:62	Complete
hme0	parfait.cs.chungbuk.ac.kr	?	Incomplete
hme0	168.188.129.123	00:60:08:a7:47:7a	Complete
hme0	168.188.129.1	00:00:A2:CB:2A:F1	Complete

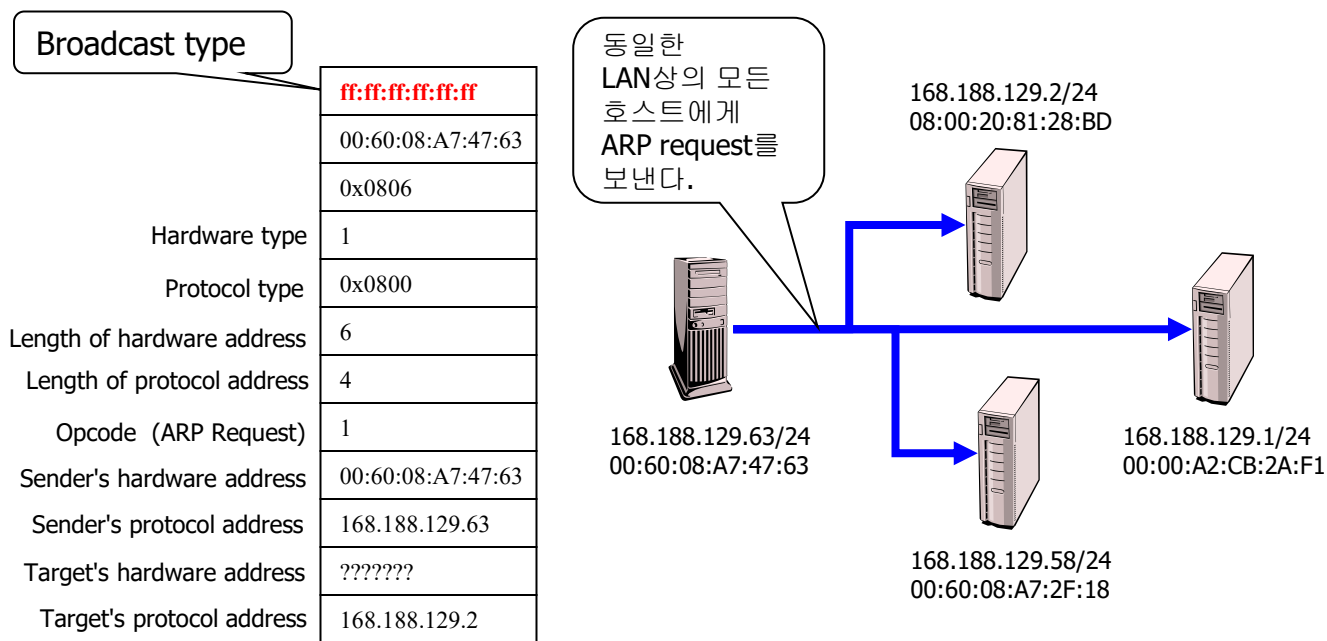
If not found Ethernet Address matching with IP Address in Cache, then Execute ARP Request/Reply Operation





Example

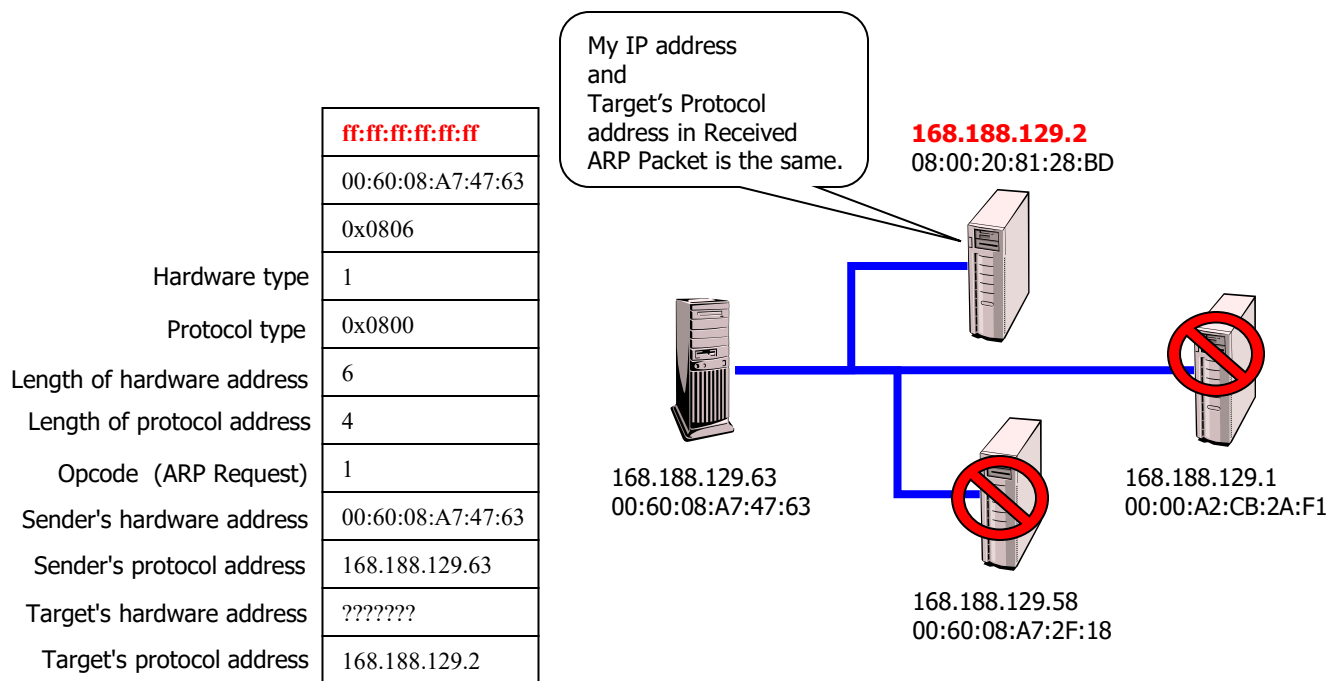
2. Processing of ARP Request in 168.188.129.63





Example

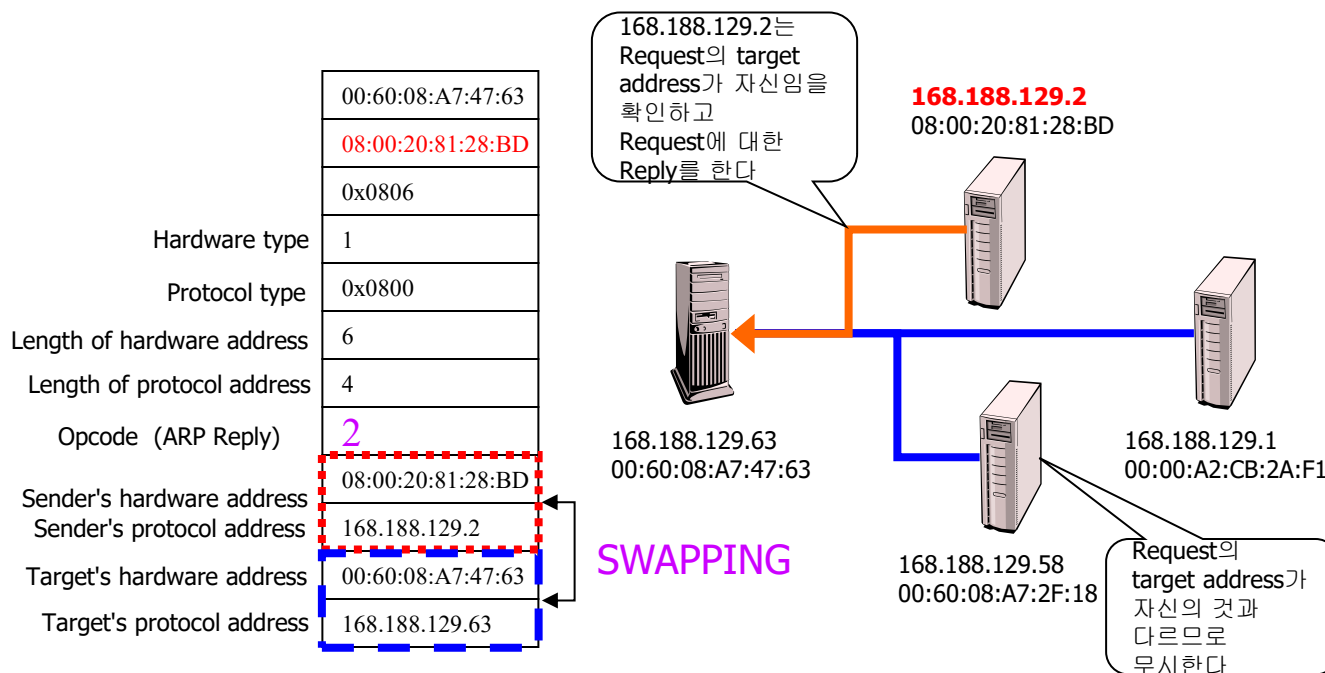
3. Processing of ARP Request in 168.188.129.2





Example

4. Processing of ARP Reply in 168.188.129.2





Example

5. Append Entry to ARP Cache in 168.188.129.63

Interface	IP Address	Ethernet Address	Status
hme0	java.cs.chungbuk.ac.kr	00:60:08:a7:48:7d	Complete
hme0	168.188.129.126	00:60:08:a7:48:62	Complete
hme0	parfait.cs.chungbuk.ac.kr	00:60:08:a7:4b:72	Complete
hme0	168.188.129.123	00:60:08:a7:47:7a	Complete
hme0	168.188.129.1	00:00:A2:CB:2A:F1	Complete
hme0	168.188.129.2	08:00:20:81:28:BD	Complete

ARP Request/reply를 통해 알아온 168.188.129.2의 Ethernet address를 테이블에 저장하고 정해진 시간 동안 이 주소를 사용하여 메시지를 전송





ARP Architecture

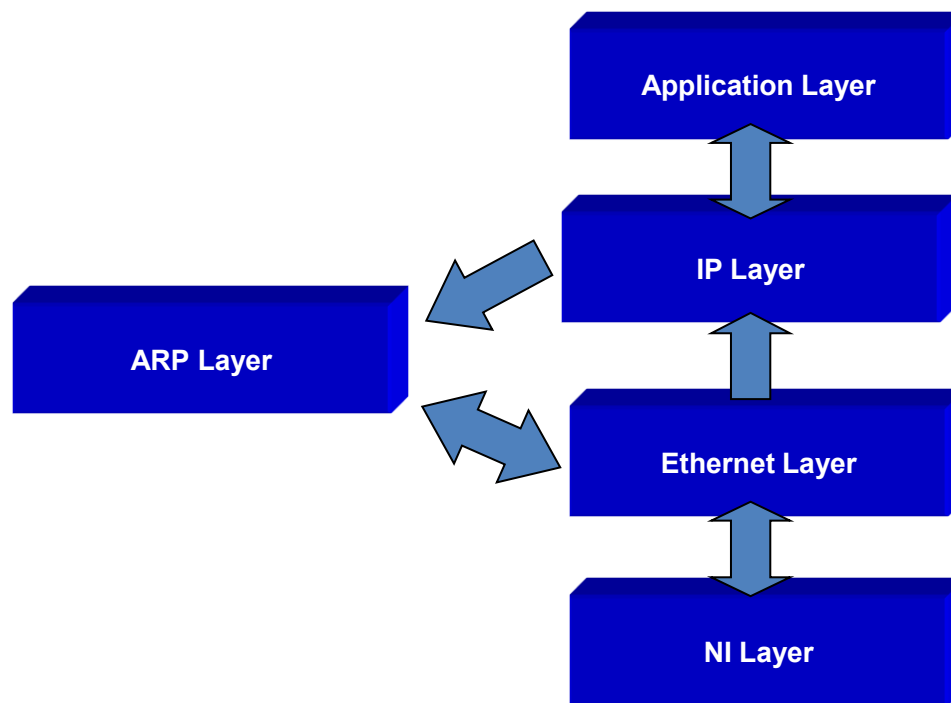
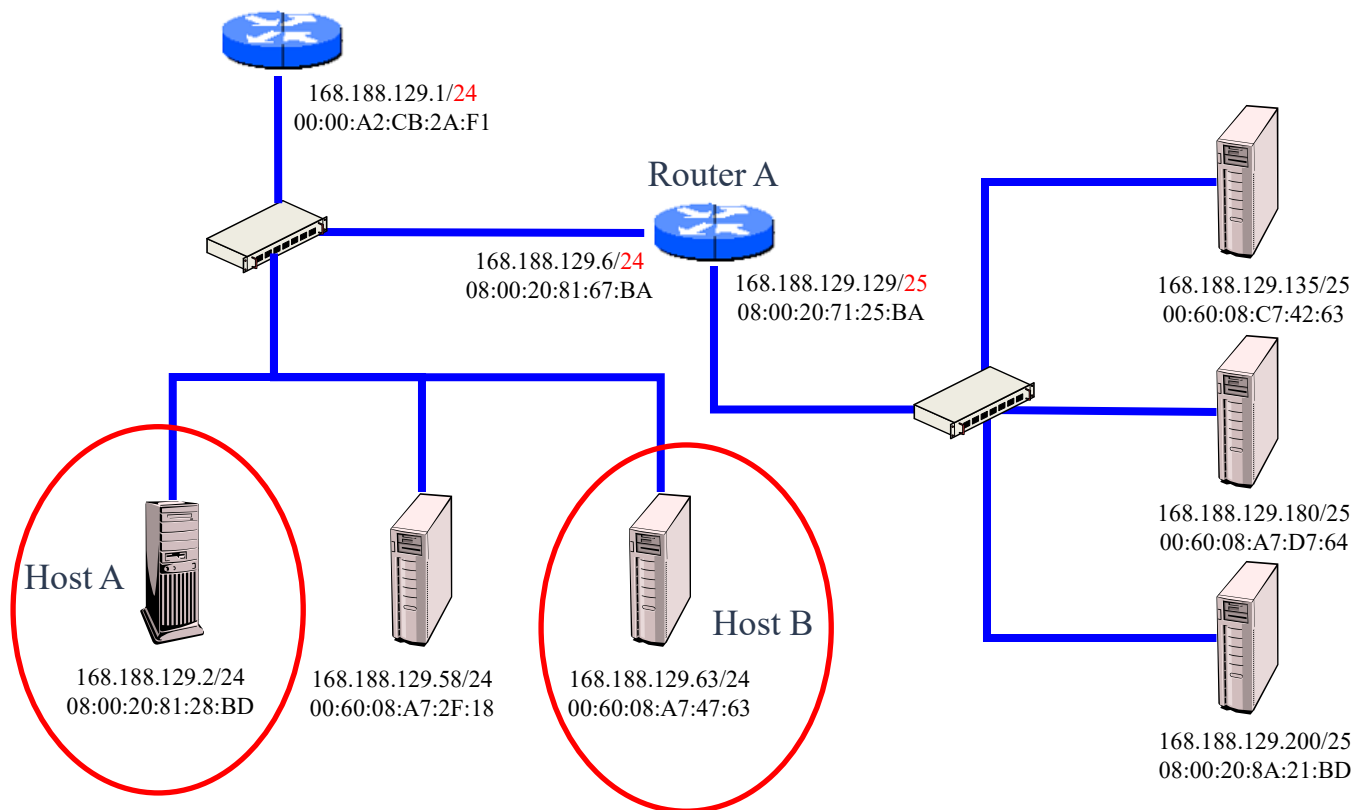


Figure 1. Layered Architecture





Network Topology for Test





Request Operation

- ◆ ARP Request Operation by User
 - ❖ Input Target's IP Address from User
 - ❖ Send ARP Request Packet to Hosts on the LAN
 - ◆ Send to Packet using Broadcasting (ff:ff:ff:ff:ff:ff)
 - ❖ Wait for ARP Reply Packet that correspond to request
- ◆ ARP Requested Operation By Other Host
 - ❖ Wait for ARP Request Packet to arrive at My Host
 - ❖ Append/Update a Entry to Cache.
 - ❖ Send to Reply Packet





Reply Operation

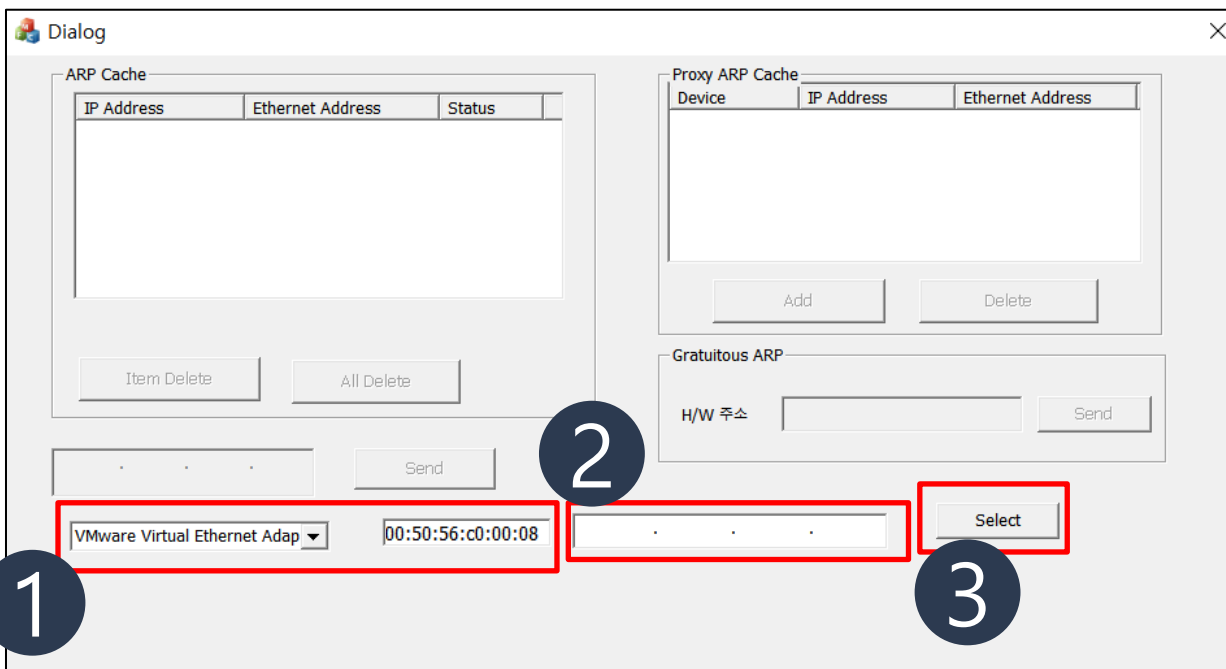
- ◆ ARP Reply Operation
 - ❖ Wait for ARP Reply Packet
 - ❖ Add/Update Cache Entry by Received ARP Reply Packet
- ◆ Return ARP Reply to Sender that response to Received ARP Requested Packet.
 - ❖ Add/Update Cache Entry by Received ARP Reply Packet





ARP Basic Operation

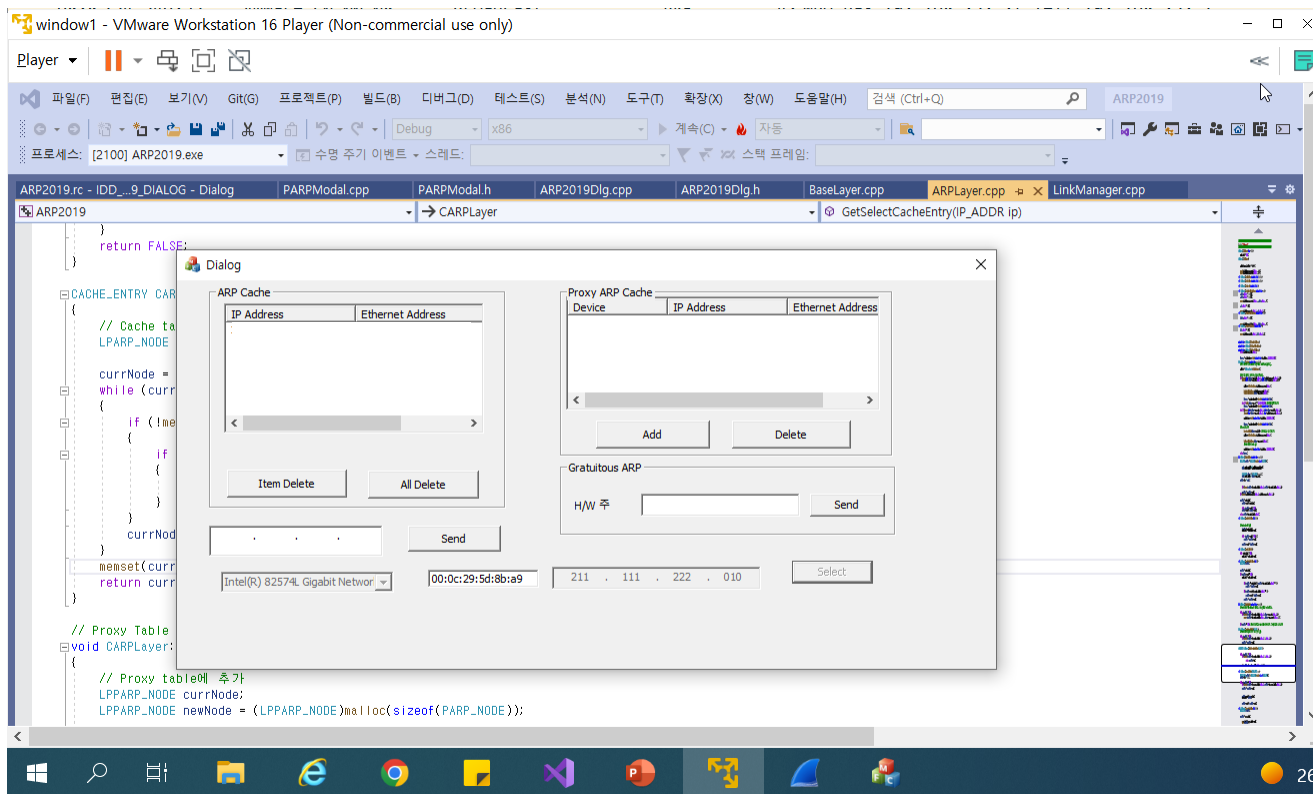
◆ Host A – Select Adapter and input IP Address





ARP Basic Operation

◆ Host B Set adapter and IP Address





ARP Basic Operation

- ◆ Host A - Send ARP Request Message to Hosts on LAN
 - ❖ IP Address: Host B's IP Address

Dialog

ARP Cache

IP Address	Ethernet Address	Status
211.111.222.10	00:00:00:00:00:00	incomplete

Item Delete All Delete

Proxy ARP Cache

Device	IP Address	Ethernet Address
--------	------------	------------------

Add Delete

Gratuitous ARP

H/W 주소 Send

211 . 111 . 222 . 010 Send

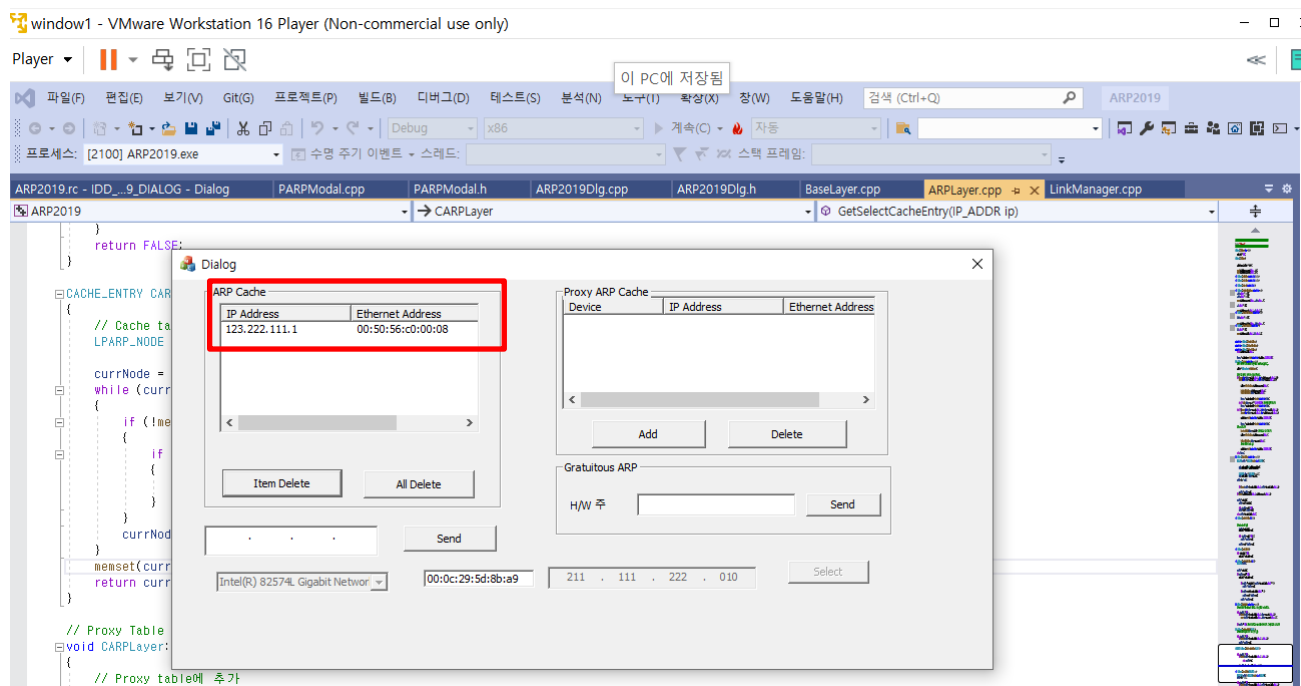
VMware Virtual Ethernet Adap 00:50:56:c0:00:08 123 . 222 . 111 . 001 Select





ARP Basic Operation

- ◆ Host B - Receive ARP Request Message from Host A and Update ARP Cache Table and then Send ARP Reply Message to Host A





ARP Basic Operation

◆ Wireshark

❖ ARP Request

17522	193.845801	VMware_c0:00:08	Broadcast	ARP	42 Who has 211.111.222.10? Tell 123.222.111.1
> Frame 17522: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{44E6B885-38C6-4494-A0A1-EEECDA42CCA1}, id 0					
> Ethernet II, Src: VMware_c0:00:08 (00:50:56:c0:00:08), Dst: Broadcast (ff:ff:ff:ff:ff:ff)					
> Address Resolution Protocol (request)					

❖ ARP Reply

17525	194.559718	VMware_5d:8b:a9	VMware_c0:00:08	ARP	42 211.111.222.10 is at 00:0c:29:5d:8b:a9
> Frame 17525: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{44E6B885-38C6-4494-A0A1-EEECDA42CCA1}, id 0					
> Ethernet II, Src: VMware_5d:8b:a9 (00:0c:29:5d:8b:a9), Dst: VMware_c0:00:08 (00:50:56:c0:00:08)					
> Address Resolution Protocol (reply)					





ARP Basic Operation

- ◆ Host A - Receive ARP Reply Message from Host B and Update ARP Cache Table

Dialog

ARP Cache

IP Address	Ethernet Address	Status
211.111.222.10	00:0c:29:5d:8b:a9	complete

Item Delete All Delete

Proxy ARP Cache

Device	IP Address	Ethernet Address
--------	------------	------------------

Add Delete

Gratuitous ARP

H/W 주소 Send

211 . 111 . 222 . 010 Send

VMware Virtual Ethernet Adap 00:50:56:c0:00:08 123 . 222 . 111 . 001 Select



Notice

◆ 실습 목표

❖ ARP 프로그램 구현

- ◆ File Transfer 프로젝트를 활용하여 구현
- ◆ NILayer, EthernetLayer, BaseLayer, LayerManager 는 그대로 재사용
- ◆ EthernetLayer 의 Frame Type 을 0x0806 으로 설정
- ◆ ARPLayer 를 새로 작성
- ◆ 새 MFC 프로젝트로 만들어도 무방





Notice

◆ 최종 과제 제출 일

- ❖ 10월 12일 23:59까지
- ❖ 조 별 대표 1명만 제출

◆ 제출물

- ❖ 프로젝트 .cpp and .h files
- ❖ 보고서
 - ◆ 보고서 양식
 - Chatting&FileTransfer 참고

◆ 데모 예정일

- ❖ Basic ARP, PARP, GARP 프로그램 데모 예정
- ❖ 11월 17일(토)~18일(일)
 - ◆ 시간 미정



Q&A

CLab.